

By: Nouran Haitham, Malak Khaled, Malek Ahmed, AbdEl-Rhman Osama, Mohamed Wael, Mohamed Akram

Supervised by: Dr. Hanan Hindy, AL. Radwa Reda

Faculty of Computer and Information Sciences - Ain Shams University

Introduction

Public speaking is a fundamental skill for academic success and career development; however, many individuals experience speech anxiety and struggle to communicate confidently. Existing AI-based coaching systems typically evaluate isolated aspects of presentation performance, such as speech or facial expressions, while lacking a comprehensive, content-aware assessment that considers both delivery and presentation material.

The growing demand for effective communication skills in education and industry highlights the need for intelligent coaching solutions. Studies indicate that nearly 50% of university students experience high presentation anxiety, while 42% report moderate anxiety, emphasizing the importance of accessible, personalized training that extends beyond traditional human feedback.

To address these challenges, we propose Elevare, an AI-powered multimodal public speaking coach that integrates emotion recognition, presentation understanding, intelligent question generation, answer evaluation, and personalized feedback within a unified framework. By combining Computer Vision, Speech Analysis, Large Language Models, and Conversational AI, Elevare provides an interactive, content-aware coaching experience that helps users improve both presentation delivery and subject mastery.

Methods

Elevare is an end-to-end AI framework that transforms presentation videos and slides into personalized coaching through four integrated stages.

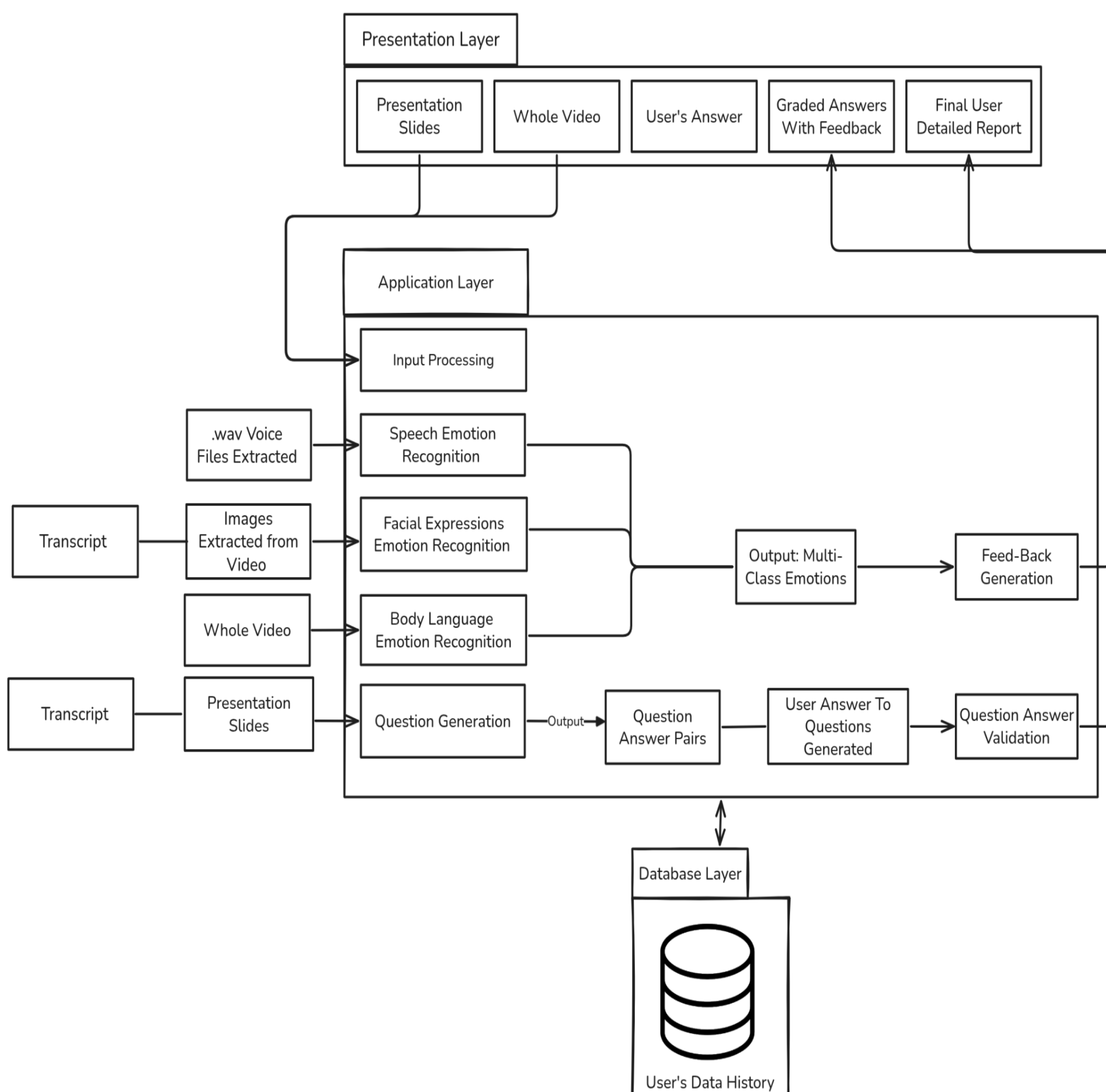


Figure 1. System Architecture

Phase 1: Multimodal Data Processing

Presentation videos are decomposed into speech, facial, and body-language streams, while presentation slides are semantically processed to prepare contextual information for subsequent AI modules.

Phase 2: Multimodal Emotion Recognition

Three specialized deep learning models independently analyze the speaker's communication channels: a Stacked Bi-LSTM for speech emotion recognition, an optimized GiMeFive CNN for facial expression analysis, and a Temporal CNN for body-language recognition. The extracted emotional cues are then synchronized to provide a comprehensive assessment of presentation delivery.

Phase 3: Intelligent Presentation Assessment

A Semantic Graph Conversational Question Generation (SG-CQG) pipeline automatically generates context-aware questions from the uploaded presentation slides. User responses are subsequently evaluated using a Retrieval-Augmented Generation (RAG) framework powered by Llama-3.3-70B, producing objective performance scores and detailed response analysis.

Phase 4: Personalized AI Coaching

A Dual-Agent Feedback Architecture combines multimodal emotion analysis with question-answer evaluation to generate personalized coaching reports. The first agent performs objective analytical reasoning, while the second synthesizes actionable recommendations, providing users with comprehensive feedback on both presentation delivery and content mastery.

Results

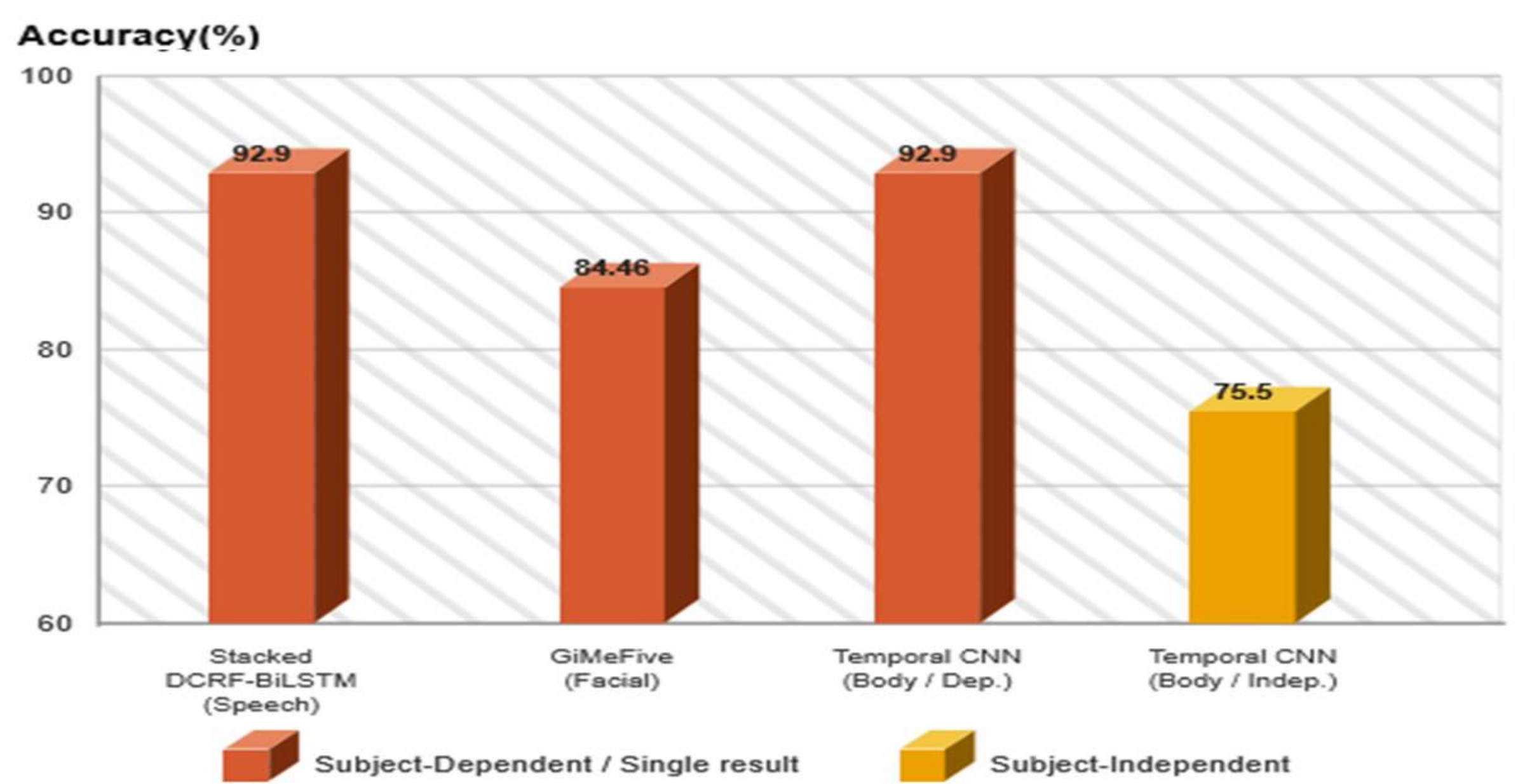


Figure 2. Summary of System Modules' Results

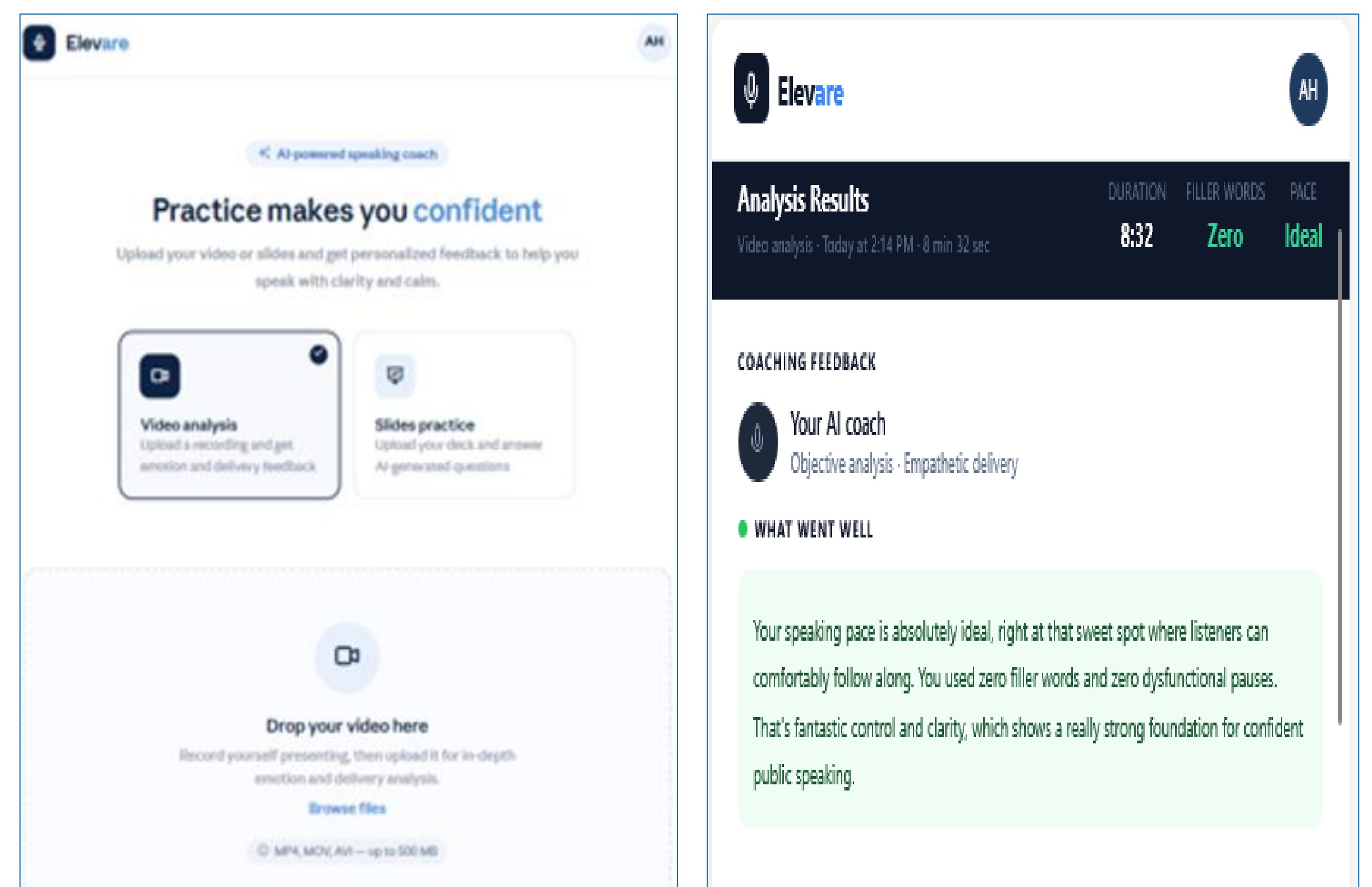


Figure 3. UI System Pages

Conclusions

Elevare introduces a unified AI-powered public speaking coach that combines multimodal emotion recognition, content-aware presentation assessment, and personalized feedback within a single framework. The proposed system achieved strong performance across its core modules (up to 92.9% emotion recognition accuracy and 72.48% answer evaluation accuracy), providing an intelligent, interactive solution that enhances both presentation delivery and subject mastery.

Bibliography

- [1] S. F. Zakaria, R. Rusli, N. H. C. Mat, and F. Tazijan, "An insight to attitudes and challenges in oral presentations among university students," in Proc. Eur. Proc. Educ. Sci., vol. 3, pp. 392–401, 2023.
- [2] F. Yaşar and S. A. Alkaya, "The effect of communication skills training on self-esteem and assertiveness in nursing students," Nurse Education Today, vol. 52, pp. 169–174, May 2017, doi: 10.1016/j.nedt.2017.02.004